Transactions of the Royal Society of Tropical Medicine and Hygiene (2005) **99**, 525—531





www.elsevierhealth.com/journals/trst

Identification of three members of the *Anopheles funestus* (Diptera: Culicidae) group and their role in malaria transmission in two ecological zones in Nigeria

T.S. Awolola[a,d,*], I.O. Oyewole[b], L.L. Koekemoer[c,d], M. Coetzee[c,d]

[a] *Molecular Entomology/Parasitology Research Laboratory, Public Health Division, Nigerian Institute of Medical Research, PMB. 2013, Yaba, Lagos, Nigeria*
[b] *Department of Basic and Applied Sciences, Babcock University, Ogun State, Nigeria*
[c] *Medical Entomology, Department of Clinical Microbiology and Infectious Diseases, School of Pathology of the National Health Laboratory Services and the University of the Witwatersrand, Johannesburg, South Africa*
[d] *Vector Control Reference Unit, National Institute for Communicable Diseases, P.O. Box 1038, Johannesburg 2000, South Africa*

Received 30 April 2004; received in revised form 3 December 2004; accepted 3 December 2004

KEYWORDS
Malaria;
Anopheles funestus;
Anopheles rivulorum;
Anopheles leesoni;
PCR;
Nigeria

Summary The role of the *Anopheles funestus* group in malaria transmission was investigated in two ecological zones in Nigeria. Sampling was carried out at four sites each around Ibadan (forest) and Ilorin (savanna). Human landing catches were supplemented with indoor and outdoor resting collections. PCR was used to identify 1848 *A. funestus* group mosquitoes to species level (749 in the savanna, 1099 in the forest) and three species were identified. In the forest, *A. funestus* s.s. predominated (55.4%), followed by *A. rivulorum* (27.6%) and *A. leesoni* (17.0%). *Anopheles funestus* was found mostly indoors. *Anopheles rivulorum* and *A. leesoni* predominated in outdoor collections ($P < 0.001$). Only *Anopheles funestus* s.s. was found in the savanna. ELISA analysis of 803 blood meal-positive specimens showed that over half of the blood meals were taken from humans in both ecotypes. The human blood index in *A. funestus* from the two study areas was similar. *Anopheles funestus* s.s. was the only species found positive for *Plasmodium falciparum* using ELISA, with overall infection rates of 2.3% and 1.0% in the forest and savanna respectively. The presence of three *A. funestus* species in Nigeria emphasizes the desirability of correct species identification within a malaria vector control programme.
© 2005 Royal Society of Tropical Medicine and Hygiene. Published by Elsevier Ltd. All rights reserved.

* Corresponding author. Tel.: +234 1 815 2803; fax: +234 1 342 5171.
E-mail address: awololas@hotmail.com (T.S. Awolola).

0035-9203/$ — see front matter © 2005 Royal Society of Tropical Medicine and Hygiene. Published by Elsevier Ltd. All rights reserved.
doi:10.1016/j.trstmh.2004.12.003

1. Introduction

Anopheles funestus Giles is a major vector of malaria in Africa, second only to *A. gambiae* Giles (De Meillon et al., 1977; Gillies and De Meillon, 1968). It belongs to a group of nine morphologically similar species that can be identified only at certain stages of their development (Gillies and Coetzee, 1987; Gillies and De Meillon, 1968). Three members of the group are virtually indistinguishable at all stages of the life cycle (*A. funestus, A. vaneedeni* Gillies & Coetzee and *A. parensis* Gillies). Two are similar to *A. funestus* in the adult stage but their immature stages are unknown (*A. aruni* Sobti and *A. fuscivenosus* Leeson). Two are unique in larval and egg characteristics (*A. confusus* Evans & Leeson and *A. leesoni* Evans), and two are different to the others by larval morphology but virtually identical to each other (*A. rivulorum* Leeson and *A. brucei* Service). Until recently only *A. funestus* was implicated in malaria transmission in Africa, but Wilkes et al. (1996) have shown that *A. rivulorum* also has a minor role in Tanzania. In most parts of Africa, *A. funestus* occurs in sympatry with other members of the group, such *A. parensis, A. rivulorum* and *A. leesoni* (Gillies and Coetzee, 1987; Gillies and De Meillon, 1968; Kamau et al., 2003).

Morphological and cytogenetic methods have been the conventional methods for distinguishing members of the *A. funestus* group (De Meillon et al., 1977; Green and Hunt, 1980; Kamau et al., 2003). However, overlapping morphological characters lead to misidentification of some species. Cytogenetic determination requires specific life stages, i.e. half-gravid females or fourth instar larvae, and *A. funestus* and *A. vaneedeni* remain indistinguishable except for a single polymorphic inversion difference. These constraints and the need to clarify the epidemiological role of the different species led to the development of molecular assays for species identification. The first molecular assay for the *A. funestus* group was carried out by Koekemoer et al. (1999) to identify four members of the group. More recently an allele-specific cocktail PCR assay that differentiates the five most common members of the group has been developed (Koekemoer et al., 2002a) and evaluated (Weeto et al., 2004). This assay is based on the same PCR principle used for the identification of members of the *A. gambiae* complex (Scott et al., 1993).

In Nigeria, a considerable number of studies have been carried out on adult anophelines but only two studies have investigated the biology of *A. funestus*. Early field observations during two malaria control projects, one in the north (Bruce-Chwatt et al., 1957) and the other in the south (Service, 1960), gave indications of the possibility of two different strains of *A. funestus*. However, in both studies, specimens collected were not identified to species level due to the difficulties in the morphological identification techniques. As a result, the species composition and distribution of the *A. funestus* group is still uncertain in Nigeria.

In the present study, two PCR assays were used to identify the different members of the *A. funestus* group, and ELISA tests were used to study their role in malaria transmission in two ecological zones where little or no information was available.

2. Materials and methods

2.1. Study areas

The study was carried out around Ibadan (03°54′E, 07°26′N), a forest area of southern Nigeria and Ilorin (04°35′E, 08°29′N), a Guinea savanna area of north central Nigeria where preliminary entomological data revealed *A. funestus* to be an important malaria vector (Awolola et al., 2003). Four study sites were selected in each of the areas, based on the availability of permanent, slow-moving water bodies that could serve as breeding sites for *A. funestus*.

The people in the study sites around Ibadan are peasant farmers cultivating maize, manioc and small cocoa plantations, which are close to the communities. Residential houses are mostly western type (cement bricks and corrugated iron roofs). Goats and sheep are common and kept in the dwellings inhabited by the people. Nomadic Fulani herdsmen occasionally bring cattle close to the human communities. Each of the study sites lacks basic infrastructure such as piped water and waste disposal facilities but has its own primary health care clinic, primary school and electricity through the rural electrification scheme of the State Government.

The main occupations in the study sites around Ilorin are subsistence farming and livestock herding. Here, cattle are kept in outdoor enclosures close to human dwellings. The chief crops are maize, cotton, groundnuts and millet. The vegetation is best described as derived savanna as little forest is left as a result of continuous deforestation. Residential houses are traditional (mud walls) with a few of western type.

The climate in Nigeria is tropical and as one moves from the south to the north of the country, the mean annual rainfall decreases from >2400 mm

in the rainforest area to <450 mm in the Sahel savanna with a marked dry season ranging from 5 to 9 months across the rainforest to the savanna belts (Oboli and Harrison, 1961).

People in both areas are quite aware of the role of mosquitoes in malaria transmission. Before the start of the project, their consent was obtained to allow the research team access to their houses. Only houses where such approval was given were used during sample collection. The study received ethical clearance from the ethical committee of the Nigerian Institute of Medical Research, Lagos, Nigeria.

2.2. Mosquito collection and processing

Adult anophelines were sampled quarterly at four sites in the forest and four sites in the savanna ecotype from October 2002 to September 2003, using volunteer human landing catches inside and outside human dwellings from 19:00 to 06:00 hours, supplemented with daytime indoor resting collections from two rooms which were different from those used for human bait collection. Human baits were offered malaria prophylaxis for the duration of the study. Outdoor resting collections were made from artificial pit shelters using standard mosquito collection techniques (Service, 1977). Collection at each site lasted for two days and each mosquito collected was anaesthetized with chloroform vapour and preserved individually on desiccated silica gel. In the laboratory, *Anopheles* mosquitoes were identified as far as possible using the morphological keys of Gillies and De Meillon (1968) and Gillies and Coetzee (1987). Only *A. funestus* group mosquitoes were included in the final analysis.

2.3. PCR identification of *Anopheles funestus*

Mosquito specimens belonging to the *A. funestus* group were analysed using two independent PCR assays. First the rDNA internal transcribed spacer 2 (ITS2) sequence variation (Hackett et al., 2000) was used on all mosquito specimens to detect *A. funestus* and *A. rivulorum*, the only members of the group that have so far been implicated in malaria transmission. Further identification was carried out using the multiplex PCR assay developed by Koekemoer et al. (2002a). In each case, the amplified DNA was separated on a 2.5% agarose gel stained with ethidium bromide and visualized on a UV transilluminator. Standard specimens (positive controls) of *A. funestus, A. rivulorum* and *A. leesoni* were provided by the Vector Control Reference Unit of the National Institute for Communicable Diseases, Johannesburg, South Africa.

2.4. ELISA

ELISA tests for *Plasmodium falciparum, P. malariae* and *P. ovale* sporozoite infections were carried out on 1637 PCR-identified female mosquitoes. The head and thorax of each mosquito was crushed in PBS (pH 7.4) and tested for circumsporozoite antigen using a double antibody ELISA (Burkot et al., 1984). Results were first scored visually and then photometrically at 405 nm using an ELISA plate reader (Anthos 2010, Anthos Labtec GmbH, 5071 Wals/Salzburg, Austria) 30 min after the substrate had been added (Beier et al., 1990). To avoid false positive ELISA readings, repeat tests were done on all sporozoite-positive samples and the cut-off point for positive tests increased to three times the mean of the negative control. Mosquitoes from indoor or outdoor resting collections and those that appeared to be blood-fed from human landing catches were analysed for their blood meal source using direct ELISA (Beier et al., 1988).

2.5. Statistical analysis

The mosquito populations from each of the study areas were analysed separately. Differences between study areas, collection methods and mosquito species collected were analysed using the χ^2 test.

3. Results

3.1. Species composition of *Anopheles funestus* in the study areas

A total of 1905 *A. funestus* group mosquitoes were characterized morphologically from the two study ecotypes of which 1848 (97.0%) specimens were identified to species level by PCR (Table 1). The results of the cocktail PCR assay revealed three different species in the forest ecotype, identified by characteristic bands of 505 bp band associated with *A. funestus* s.s., 421 bp with *A. rivulorum* and 146 bp with A. *leesoni* (Figure 1). In the forest ecotype, *A. funestus* comprised 55.4% of the sample, *A. rivulorum* 27.6% and *A. leesoni* 17.0% (Table 1). *Anopheles funestus* was mostly found indoors, while *A. rivulorum* and *A. leesoni* were predominant in the outdoor collections. *Anopheles funestus* was the only species found in the savanna

Table 1 Species composition of *Anopheles funestus* group mosquitoes caught indoors or outdoors in forest and savanna ecotypes, Nigeria

Species	Ecotype		
	Indoor collection	Outdoor collection	Total (%) caught per ecotype
Forest			
A. funestus	418	191	609 (55.4)
A. rivulorum	53	250	303 (27.6)
A. leesoni	73	114	187 (17.0)
Savanna			
A. funestus	599	150	749 (100)
Total	1143	705	1848

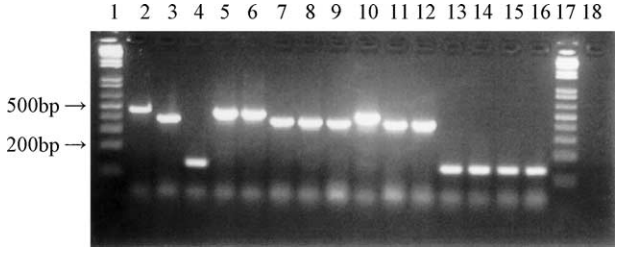


Figure 1 PCR identification of three species within the *Anopheles funestus* group. Lanes 1 and 17: 1 kb DNA ladder (standard); lane 2: positive control (*A. funestus* s.s.); lane 3: positive control (*A. rivulorum*); lane 4: positive control (*A. leesoni*); lane 18: negative control; lanes 5-16: sample templates, *A. funestus* s.s. (lanes 5, 6 and 10), *A. rivulorum* (lanes 7—9 and 11—12) and *A, leesoni* (lanes 13—16).

ecotype where 80.0% were caught resting or on human bait indoors (Table 1).

3.2. Blood meal sources

A total of 803 (43.5%) of the 1848 mosquitoes identified by PCR were blood-fed, 454 from the forest ecotype and 349 from the savanna ecotype. Overall, more than half of the blood-fed mosquitoes investigated in the two study ecotypes were positive for human blood meal, 16.9% were positive for bovine blood meal, 19.6% were positive for mixed blood meal (human and bovine), and 8.5% were scored as negative for both human and bovine blood meal and presumably had fed on other hosts (Table 2). *Anopheles funestus* had a significant preference for human blood meal only ($\chi^2 = 89.9$, $d.f. = 3$, $P < 0.001$) compared with *A. rivulorum* and *A. leesoni*. None of the *A. leesoni* had fed exclusively on humans (Table 2).

The proportions of blood meals of *A. funestus* on human, bovine and both hosts were fairly similar in the two ecotypes.

3.3. *Plasmodium falciparum* sporozoite rates

Overall, 2.3% (12/530) and 1.0% (7/682) of the *A. funestus* tested in the forest and savanna ecotypes respectively were positive by ELISA for *P. falciparum* infection and none were positive for *P. malariae* or *P. ovale*. Although the sporozoite rate

Table 2 Number of blood-fed *Anopheles funestus* group mosquitoes caught indoors or outdoors found to contain blood from humans and cattle hosts in forest and savanna ecotypes, Nigeria

Ecotype Species	*n*	Indoors *n* (%)	Outdoors *n* (%)	Human only *n* (%)	Cattle only *n* (%)	Human and cattle *n* (%)	Negative for human and cattle *n* (%)
Forest							
A. funestus	314	264 (84.1)	50 (15.9)	207 (65.9)	20 (6.4)	62 (19.7)	25 (8.0)
A. rivulorum	106	22 (20.8)	84 (79.2)	16 (15.1)	61 (57.5)	26 (24.5)	3 (2.8)
A. leesoni	34	11 (32.4)	23 (67.6)	0	20 (58.8)	9 (26.5)	5 (14.7)
Savanna							
A. funestus	349	299 (85.7)	50 (14.3)	219 (62.8)	35 (10.0)	60 (17.2)	35 (10.0)
Total	803	596 (74.2)	207 (25.8)	442 (55.0)	136 (16.9)	157 (19.6)	68 (8.5)

Table 3 Infection in *Anopheles funestus* group mosquitoes tested with *Ptasmodium falciparum* circumsporozoite antigen from indoor and outdoor collections in forest and savanna ecotypes, Nigeria

Ecotype Species	Total no. examined		No. (%) positive		
	Indoors	Outdoors	Indoors	Outdoors	Total
Forest					
A. funestus	400	130	10 (2.5)	2 (1.5)	12 (2.3)
A. rivulorum	53	211	0	0	0
A. leesoni	60	101	0	0	0
Savanna					
A. funestus	562	120	6 (1.1)	1 (0.8)	7 (1.0)

in *A. funestus* in indoor and outdoor collections combined appeared higher in the forest ecotype compared with the savanna ecotype, the difference was not statistically significant ($\chi^2 = 3.2$, $d.f. = 1$, $P > 0.05$). Infection rates in mosquitoes from the different methods of collection in both study areas were also similar ($P = 0.06$ by Fisher's exact test). None of the *A. rivulorum* or *A. leesoni* tested positive for *P. falciparum* infection (Table 3).

4. Discussion

In Nigeria, malaria vector control programmes have to deal with the different *Anopheles* species involved in malaria transmission. Although most of the information available is on the *A. gambiae* complex, there is a renewed interest in the study of the *A. funestus* group due to the availability of molecular assays that allow easy and rapid identification of the species in this group. The present study is the first report on the molecular analysis of the *A. funestus* group in Nigeria and has established the presence of *A. funestus, A. rivulorum* and *A. leesoni*. Although Hackett et al. (2000) used the ITS2 size variation to differentiate between *A. funestus* and *A. rivulorum*, a recent report demonstrated overlap between *A. funestus* and *A, vaneedeni* (~850 bp) and between *A. rivulorum* and *A. leesoni* (~520 bp) (Koekemoer et al., 2002b). Here, we circumvented this problem by using the multiplex PCR assay (Koekemoer et al., 2002a) that can identify the five most common members of the group. About 3.0% of the specimens recorded as *A. funestus* using morphological keys could not be identified by PCR. This could be due to misidentification of some of the specimens, problems associated with DNA extraction and amplification, or the presence of other members of the *A. funestus* group which could not be identified by the PCR assays but require further analysis.

The results of the ELISA-based investigation of blood meal sources indicate that *A. funestus* is predominantly anthropophagic in both ecotypes, whereas *A. rivulorum* and *A. leesoni* fed mainly on cattle (none of the *A. leesoni* had fed exclusively on humans) in the forest ecotype. This is similar to previous results where *A. leesoni* was found to feed mainly on cattle even in the presence of human hosts (Gillies and De Meillon, 1968). *Anopheles funestus* was the only species found to be infected with *P. falciparum* circumsporozoites. This supports earlier studies that showed that *A. leesoni* and *A. rivulorum* do not play a role in malaria transmission in West Africa (Boccolini et al., 1994; Costantini et al., 1999). A recent study revealed false ELISA results in *A. marshallii* in South Africa when the cut-off point for positive ELISA results was taken as twice the mean of the negative control (Koekemoer et al., 2001). Here, in addition to repeated tests done on positive specimens, we reduced the chances of a false positive ELISA by increasing the cut-off point to three times the mean of the negative controls. This may well have reduced the number of mosquitoes detected as infected in our sampled population giving an underestimation of the infectivity rate. Nevertheless, the sporozoite rate recorded in the forest ecotype is comparable to that reported in a similar ecotype in West Africa (Boccolini et al., 1994), but higher than those reported in northern Nigeria (Molineaux and Gramiccia, 1980). However, none of the earlier studies distinguished the different members of the *A. funestus* group, and this could have reduced the infection rates in *A. funestus*, assuming that *A. funestus* constituted only a proportion of the *A. funestus* group collected.

During the study, a number of other anophelines were encountered, including *A. gambiae* s.s., *A. arabiensis, A. nili, A. pharoensis, A. coustani* and *A. hancocki*. The first three species have been identified as malaria vectors in the area (Awolola et al.,

2003). The fact that only *A. funestus* within the *A. funestus* group was found infected to be with *P. falciparum* underlines the need for correct identification of malaria vectors that belong to species complexes or groups. Failure to recognize sibling species could mean that vector species are mistaken for non-vectors. This could be misleading for vector control programmes if morphologically defined species are actually a mixture of two or more species whose behaviour may differ depending on the ecotype or their vectorial capacity.

Conflicts of interest statement
The authors have no conflicts of interest concerning the work reported in this paper.

Acknowledgements

We thank our field assistants for painstaking efforts during the field activities. We appreciate the support of officers of the Malaria Control Programme at Egbeda for facilitating the field activities. Reagents for sporozoite detections were provided by the Centers for Disease Control and Prevention, Atlanta, GA, USA, through Dr R. Wirtz. This investigation received financial support under the Multilateral Initiative on Malaria (MIM) project A30026 through the UNICEF/UNDP/World Bank/WHO Special Programme for Research and Training in Tropical Diseases (TDR) and the Wellcome Trust through grant number GR070472 to M.C. and T.S.A.

References

Awolola, T.S., Ibrahim, K., Okorie, T., Hunt, R.H., Coetzee, M., 2003. Species composition and biting activities of anthropophilic *Anopheles* mosquitoes and their role in malaria transmission in a holoendemic area in south-western Nigeria. Afr. Entomol. 11, 78—82.

Beier, J.C., Perkins, P.V., Wirtz, R.A., Koros, J., Diggs, D., Gargan, T.P., Koech, D.K., 1988. Blood meal identification by direct enzyme-linked immunosorbent assay (ELISA), tested on *Anopheles* (Diptera: Culicidae) in Kenya. J. Med. Entomol. 25, 9—16.

Beier, J.C., Perkins, P.V., Koros, J.K., Onyango, F.K., Gargan, T.P., Wirtz, R.A., Koech, D.K., Robert, C.R., 1990. Malaria sporozoite detection by dissection and ELISA to assess infectivity of Afrotropical *Anopheles* (Diptera: Culicidae). J. Med. Entomol. 27, 377—384.

Boccolini, D., Sabatini, A., Sanogo, E., Sagnon, N.F., Coluzzi, M., Costantini, C., 1994. Chromosomal and vectorial heterogeneities in *Anopheles funestus* from Burkina Faso, West Africa. Parassitologia 36, 1—20.

Bruce-Chwatt, L.J., Archibald, H.M., Haworth, J., 1957. Malaria control in pilot project in Western Sokoto, northern Nigeria. Annual Report, Ministry of Health Northern Nigeria 3, 1—55.

Burkot, T.R., Williams, J.L., Schneider, I., 1984. Identification of *Plasmodium falciparum*-infected mosquitoes by a double antibody enzyme-linked immunosorbent assay. Am. J. Trop. Med. Hyg. 33, 783—788.

Costantini, C., Sagnon, N.F., Sanogo, E., Coluzzi, M., Boccolini, D., 1999. Chromosomal and bionomic heterogeneities suggest incipient speciation in *Anopheles funestus* from Burkina Faso. Parassitologia 41, 595—611.

De Meillon, B., Van Eeden, G.J., Coetzee, L., Coetzee, M., Meiswinkel, R., Du Toit, C.L.N., Hansford, C.F., 1977. Observations on a species of the *Anopheles funestus* subgroup, a suspected exophilic vector of malaria parasites in north-eastern Transvaal, South Africa. Mosq. News 37, 657—661.

Gillies, M.T., Coetzee, M., 1987. A Supplement to the Anophelinae of Africa South of the Sahara (Afrotropical Region). Publ. S. Afr. Inst. Med. Res. no. 55.

Gillies, M.T., De Meillon, B., 1968. The Anophelinae of Africa South of the Sahara (Ethiopian Zoogeographical Region). Publ. S. Afr. Inst. Med. Res. no. 54.

Green, C.A., Hunt, R.H., 1980. Interpretation of variation in ovarian polytene chromosomes of *Anopheles funestus* Giles, *Anopheles parensis* Gillies and *Anopheles aruni*? Genetica 51, 187—195.

Hackett, B.J., Gimnig, J., Guelbeogo, W., Costantini, C., Koekemoer, L.L., Coetzee, M., Collins, F.H., Besansky, N.J., 2000. Ribosomal DNA internal transcribed spacer (ITS2) sequences differentiate *Anopheles funestus* and *An. rivulorum*, and uncover a cryptic taxon. Insect Mol. Biol. 9, 369—374.

Kamau, L., Munyekenye, G.O., Koekemoer, L.L., Hunt, R.H., Coetzee, M., 2003. A survey of the *Anopheles funestus* (Diptera: Culicidae) group of mosquitoes from ten sites in Kenya with special emphasis on population genetic structure based on chromosomal inversion karyotypes. J. Med. Entomol. 40, 664—671.

Koekemoer, L.L., Lochouarn, L., Hunt, R.H., Coetzee, M., 1999. Single-strand conformation polymorphism analysis for identification of four members of the *Anopheles funestus* (Diptera: Culicidae) group. J. Med. Entomol. 36, 125—130.

Koekemoer, L.L., Rankoe, E.M., La Grange, J.P., Govere, J., Coetzee, M., 2001. False detection of *Plasmodium falciparum* sporozoite in *Anopheles marshallii* group of mosquitoes. J. Am. Mosq. Cont. Assoc. 17, 160—165.

Koekemoer, L.L., Kamau, L., Hunt, R.H., Coetzee, M., 2002a. A cocktail polymerase chain reaction (PCR) assay to identify members of the *Anopheles funestus* (Diptera: Culicidae) group. Am. J. Trop. Med. Hyg. 66, 804—811.

Koekemoer, L.L., Hargreaves, K., Hunt, R.H., Coetzee, M., 2002b. Identification of *Anopheles parensis* (Diptera:Culicidae) using ribosomal DNA internal transcribed spacer (ITS2) sequence variation. Afr. Entomol. 10, 235—239.

Molineaux, L., Gramiccia, G., 1980. The Garki Project. In: Research on the Epidemiology and Control of Malaria in the Sudan Savannah of West Africa. World Health Organization, Geneva.

Oboli, H.O.N., Harrison, R.J., 1961. An Outline Geography of West Africa, fourth ed. George G. Harrap Publishing, London.

Scott, J.A., Brogdon, W.G., Collins, F.H., 1993. Identification of single specimens of the *Anopheles gambiae* complex by polymerase chain reaction. Am. J. Trop. Med. Hyg. 49, 520—529.

Service, M.W., 1960. A taxonomic study of *Anopheles funestus funestus* Giles (Diptera: Culicidae) from southern and northern Nigeria, with notes on its varieties and synonyms. Proc. R. Entomol. Soc. Lond. 29, 77—84.

Service, M.W., 1977. A critical review of procedures for sampling populations of adult mosquitoes. Bull. Entomol. Res. 67, 343—382.

Weeto, M.M., Koekemoer, L.L., Kamau, L., Hunt, R.H., Coetzee, M., 2004. Evaluation of a species specific PCR assay for the *Anopheles funestus* group from eleven African countries and Madagascar. Trans. R. Soc. Trop. Med. Hyg. 98, 142—147.

Wilkes, T.J., Matola, Y.G., Charlwood, J.D., 1996. *Anopheles rivulorum*, a vector of human malaria in Africa. Med. Vet. Entomol. 10, 108—110.

Available online at www.sciencedirect.com

SCIENCE DIRECT®